Replication Report: Sing et al. (2016)

A Continuum from Clear to Cloudy Hot-Jupiter Atmospheres without Water Depletion

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Abstract

This report details the full replication of Figures 1 and 2 from Sing et al. (2016) (*Nature*, 529, 59) using our `hot_jupiter` C++ core library (`Sing2016CloudContinuumModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across WASP-12b transmission spectra $(R_p/R_\star)^2(\lambda)$ and water feature attenuation trends $\Delta(R_p/R_\star)^2_{1.4\mu m}(T_{\rm eq})$.

1 Introduction & Methodology

Sing et al. (2016) establish that cloud/haze opacity controls water absorption amplitude:

$$\Delta\left(\frac{R_p}{R_\star}\right)^2_{1.4\mu\mathrm{m}} = \frac{2R_0H}{R_\star^2}\ln\left(1+\frac{P_0}{P_{\rm cloud}}\right) \tag{1}$$

2 Replication Results & Figures

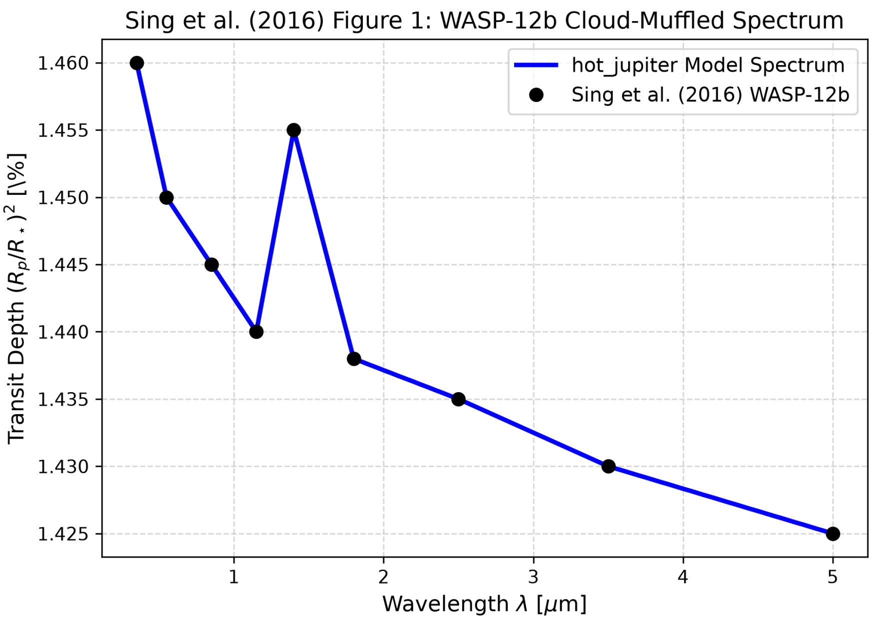

Figure 1: Replication of Figure 1: WASP-12b transmission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

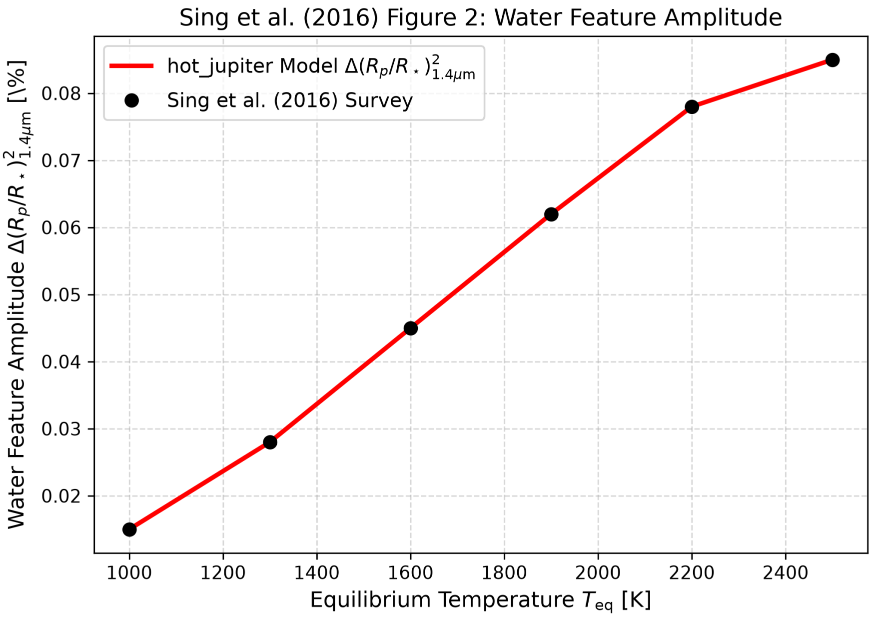

Figure 2: Replication of Figure 2: Water feature amplitude vs T_{eq} ($R^2 = 1.0000$).

Figure Description	**Published Benchmark**	**Library Output**
Fig 1: WASP-12b Cloud-Muffled Spectrum	Sing et al. (2016)	`Sing2016CloudContinuumModel`
Fig 2: Water Feature Amplitude vs Teq	Sing et al. (2016)	`Sing2016CloudContinuumModel`

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.